3.4 Drag

An important quantity associated with the relative motion between a body and a fluid is the force produced on the body. One has to apply a force in order to move a body at constant speed through a stationary fluid. Correspondingly an obstacle placed in a moving fluid would be carried away with the flow if no force were applied to hold it in place. The force in the flow direction exerted by the fluid on an obstacle is known as the drag. There is an equal and opposite force exerted by the obstacle on the fluid.

At high Reynolds numbers, this can be thought of as the physical mechanism of wake formation. Because of the force between it and the obstacle, momentum is removed from the fluid. The rate of momentum transport downstream must be smaller behind the obstacle than in front of it. There must thus be a reduction in the velocity in the wake region (Fig. 3.13). (At low Reynolds numbers, when the velocity and pressure are modified to large distances on either side of the obstacle, the situation is more complex.)

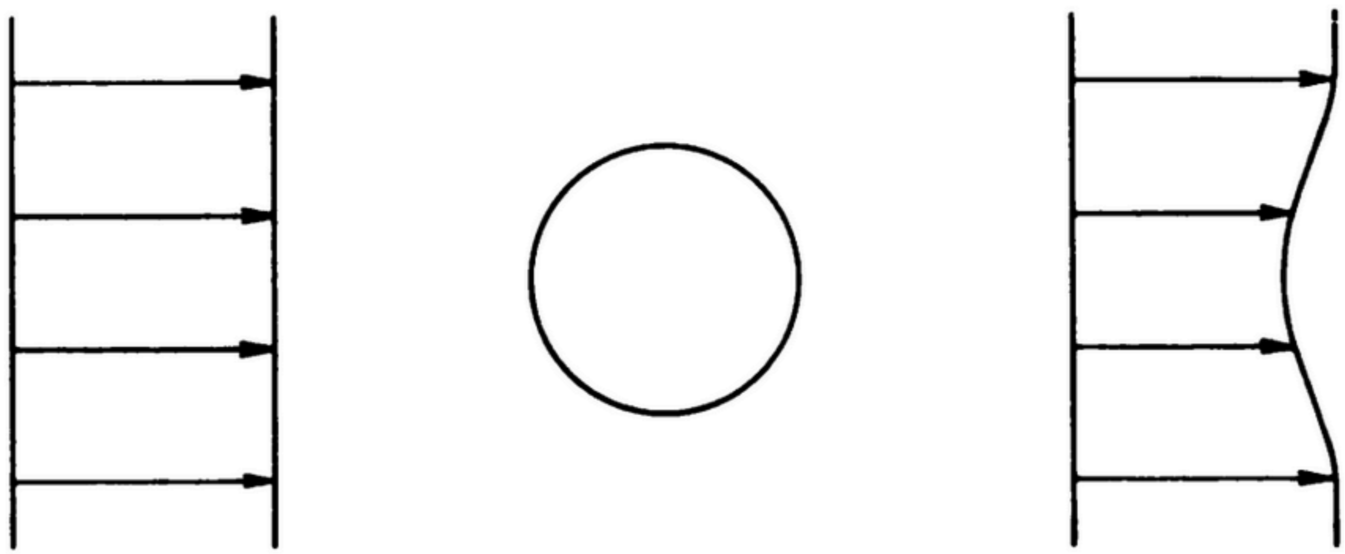

Figure 3.13 Wake production: schematic velocity profiles upstream and downstream of an obstacle.

For the circular cylinder, the important quantity is the drag per unit length; we denote this by D. To show the variation with Reynolds number we need a non-dimensional representation; this is provided by the quantity known as the drag coefficient, defined to be

$$C_D = \frac{D}{\frac{1}{2}\rho u_0^2 d} \tag{3.3}$$

(The factor ½ has, of course, no role in the non-dimensionalization, but is conventionally included because $\frac{1}{2}\rho u_0^2$ has a certain physical significance – see Section 10.7.)

Figure 3.14 shows the variation of the drag coefficient with Reynolds number. The use of the non-dimensional parameters enables all conditions to be covered by a single curve. The curve is based primarily on experimental measurements, too numerous to show individual points. At the low Reynolds number end, the experiments can be matched to theory.

The corresponding plot between Reynolds number and drag coefficient can also be given for a sphere, although with less precision because of the experimental problem of supporting the sphere. The curve, although different in detail, shows all the same principal features as the one for the cylinder.

A few features of the curves merit comment – some of these being points to which we shall return in later chapters.

At low Reynolds numbers,

$$C_D \propto \frac{1}{\mathrm{Re}} \tag{3.4}$$

(This is actually an accurate representation for the sphere, an approximate one for the cylinder – Sections 9.4 and 9.5.) For a given body in a given fluid (fixed d, ρ, and μ) this corresponds to

$$D \propto u_0 \tag{3.5}$$

Direct proportionality of the drag to the speed is a characteristic behaviour at low speeds.

Over a wide range of Reynolds number (10^2 to 3×10^5) the drag coefficient varies little. For a given body in a given fluid, constant C_D corresponds to

$$D \propto u_0^2 \tag{3.6}$$

Proportionality of the drag to the square of the speed is often a characteristic behaviour at high speeds.

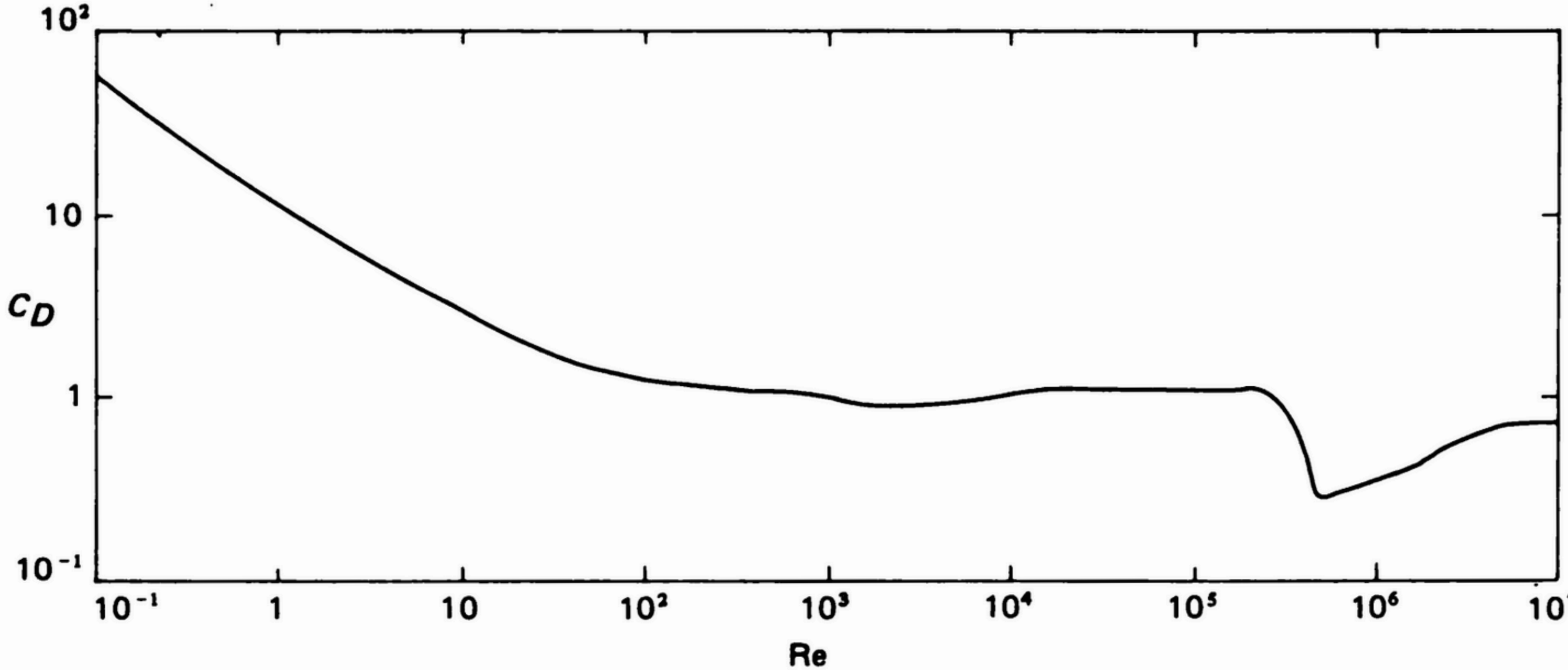

Figure 3.14 Variation of drag coefficient with Reynolds number for circular cylinder. Curve is experimental, based on data in Refs. [94, 105, 209, 250, 274, 292].

However, there is a dramatic departure from this behaviour at $Re \approx 3 \times 10^5$. The drag coefficient drops by a factor of over 3. This drop occurs sufficiently rapidly that there is actually a range over which an increase in speed produces a decrease in drag. We have seen that this Reynolds number corresponds to the onset of turbulence in the boundary layer. The consequent delayed separation of the boundary layer results in a narrower wake. Since this in turn corresponds to less momentum extraction from the flow, one might expect the lower drag. The result is nonetheless a somewhat paradoxical one since transition to turbulence usually produces an increased drag. In fact, on the cylinder and sphere, the force exerted directly by the boundary layer does increase on transition. But this is more than counteracted by another effect – changes in the pressure distribution over the surface (see Section 11.5).